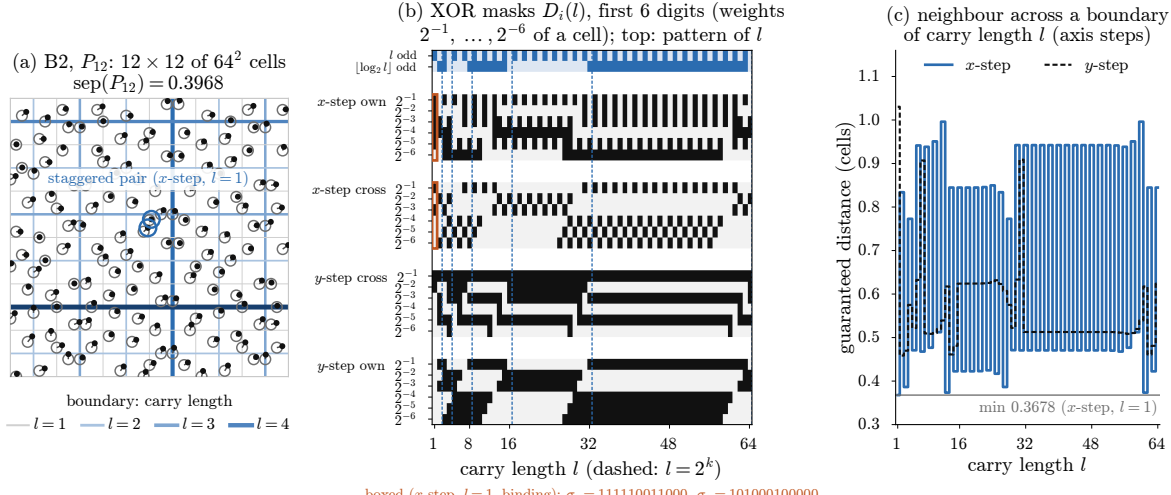
(a) B2, P_{12}: 12×12 of 64^2 cells
$\mathrm{sep}(P_{12}) = 0.3968$
staggered pair (x-step, $l = 1$)
boundary: carry length
$l = 1$
$l = 2$
$l = 3$
$l = 4$
(b) XOR masks $D_i(l)$, first 6 digits (weights $2^{-1}, \ldots, 2^{-6}$ of a cell); top: pattern of l
l odd
$\lfloor \log_2 l \rfloor$ odd
x-step own
x-step cross
y-step cross
y-step own
2^{-1}
2^{-2}
2^{-3}
2^{-4}
2^{-5}
2^{-6}
1
8
16
32
48
64
carry length l (dashed: $l = 2^k$)
boxed (x-step, $l = 1$, binding): $\sigma_0 = 111110011000$, $\sigma_1 = 101000100000$
(c) neighbour across a boundary of carry length l (axis steps)
x-step
y-step
guaranteed distance (cells)
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
min 0.3678 (x-step, $l = 1$)
1
16
32
48
64
carry length l